

UK Pharmaceutical Commercial Opportunity Analysis: SGLT2 Market

1. Project Overview

This project investigates the commercial dynamics of the SGLT2 inhibitor market in England using publicly available NHS prescribing and GP population data. The purpose of the analysis is not simply to visualise prescribing activity, but to transform raw healthcare data into decision-oriented commercial intelligence.

The analysis focuses on four major SGLT2 molecules: dapagliflozin, empagliflozin, canagliflozin, and ertugliflozin. These medicines form a useful competitive market for analysis because prescribing volumes can be observed over time and compared across products and geographies.

The project was designed from the perspective of a commercial analyst working within the pharmaceutical sector. The central question is:

Where are the strongest commercial opportunities within the SGLT2 inhibitor market in England, how is the competitive landscape changing, and which geographic areas show potential for further market development?

The analysis therefore examines market growth, competitive market share, product contribution to category expansion, geographic prescribing variation, and potential commercial whitespace.

The dashboard was developed in Power BI using Power Query for data preparation, a star-schema data model, and DAX measures for time intelligence, market-share calculations, growth decomposition, population-normalised utilisation, benchmarking, and geographic segmentation.

2. Business Context

A pharmaceutical company operating within the SGLT2 market requires more than information about how many prescriptions have been issued. Commercial decision-making also requires an understanding of how the overall therapeutic category is developing, which products are gaining or losing competitive position, and whether growth is evenly distributed across geographic markets.

Several questions therefore guide the analysis.

The first is whether the SGLT2 market is expanding and how quickly prescribing activity is increasing. The second is whether individual products are growing because they are gaining competitive share or simply benefiting from overall category expansion.

The project also investigates geographic variation. Raw prescription volumes cannot be compared directly between Integrated Care Boards because areas with larger registered populations would naturally generate greater prescribing volumes. Prescribing activity is therefore normalised using registered GP population to provide a more comparable measure of utilisation.

The ultimate objective is to move from descriptive reporting toward commercial decision support by identifying areas that may warrant further investigation or prioritisation.

3. Research Questions

The main research question for the project is:

How can publicly available prescribing data be transformed into actionable commercial intelligence for pharmaceutical portfolio decision-making?

This is supported by several analytical questions.

The analysis evaluates how the SGLT2 market has evolved over time, how the competitive position of major molecules has changed, how utilisation varies geographically across England, which geographic markets appear relatively developed or underdeveloped compared with the national benchmark, and what limitations arise when prescribing data is used as a proxy for pharmaceutical market performance.

The final question is particularly important because prescribing activity does not directly represent patients, company sales, or clinical demand. The project therefore treats commercial opportunity as an analytical screening concept rather than a definitive measure of unmet demand.

4. Data Sources

The primary dataset used in this project is the NHS Business Services Authority English Prescribing Dataset.

Monthly prescribing data from **January 2025 to June 2026** was extracted for the four SGLT2 molecules included in the analysis.

The prescribing data includes fields such as prescribing month, regional office, Integrated Care Board, GP practice, BNF chemical substance, product presentation, prescription items, quantity, Net Ingredient Cost, actual cost, and SNOMED code.

A second NHS dataset containing the number of patients registered at GP practices was incorporated to support population-normalised geographic analysis.

The registered population dataset provides a denominator that allows prescribing activity to be expressed relative to the population served by each practice rather than relying on absolute prescription volumes.

The final model therefore combines pharmaceutical prescribing activity with healthcare organisational geography and registered-patient population data.

5. Data Preparation and Transformation

The raw prescribing files were distributed across multiple monthly datasets. Power Query was used to combine the monthly extracts and create a single prescribing fact table.

One data-engineering challenge was that historical prescribing files contained slightly different field structures compared with later SNOMED-enabled files.

For example, older files used fields such as:

CHEMICAL_SUBSTANCE_BNF_DESCR

and

BNF_DESCRIPTION

while later files used fields such as:

BNF_CHEMICAL_SUBSTANCE

and

BNF_PRESENTATION_NAME.

To ensure consistency across the analytical period, these fields were harmonised into a common schema before the datasets were appended.

Standardised fields included:

YearMonth, RegionName, RegionCode, ICBName, ICBCode, PracticeName, PracticeCode, ChemicalSubstanceCode, Molecule, PresentationCode, PresentationName, Items, Quantity, NIC, ActualCost, and SNOMEDCode.

The monthly variable was also transformed from values such as 202501 into proper date values such as 01 January 2025. This allowed Power BI time-intelligence calculations to operate correctly.

A separate date dimension was subsequently created containing Date, Year, Month Number, Month Name, Quarter, and Year-Month labels.

6. Data Model

A star-schema structure was used to separate transactional prescribing data from descriptive dimensions.

The central table is:

Fact_Prescribing

This contains monthly prescribing activity at product and practice level.

Several dimension tables were created around the fact table.

Dim_Date contains the calendar structure required for monthly and year-on-year analysis.

Dim_Product contains the chemical substance code and molecule name, providing one row per SGLT2 molecule.

Dim_Presentation contains product presentation-level information, allowing analysis below the molecule level where required.

Dim_Practice contains practice-level geography, including practice, ICB, PCO, regional identifiers, and postcode information where available.

The registered population dataset was aggregated to one row per practice and connected through PracticeCode.

This structure allows descriptive fields in dimension tables to filter numerical measures stored in the prescribing fact table.

The model therefore separates analytical measures from descriptive attributes and reduces duplication within the dataset.

7. Core Measures

Several DAX measures were developed to support the analysis.

Total Prescription Items

Prescription items were selected as the primary volume measure.

The total number of prescription items is calculated as:

Total Items = SUM of prescription items

This represents prescribing volume rather than the number of individual patients.

8. Market Growth

Year-on-year growth compares the current period with the equivalent period in the previous year.

For the current dashboard, **2026 YTD refers to January to June 2026**, and growth is therefore compared with **January to June 2025**.

The market recorded approximately **20% year-on-year growth in prescription items during 2026 YTD**.

This indicates continued expansion in the SGLT2 category during the analysed period.

Importantly, the dashboard avoids comparing a full year of 2025 with only six months of 2026. Year-on-year measures use equivalent calendar periods.

9. Competitive Market Share

Market share is calculated using prescription items.

For each molecule:

Market Share = Product Prescription Items / Total SGLT2 Prescription Items

The 2026 YTD analysis indicates that **dapagliflozin is the leading molecule**, accounting for **approximately 73% of SGLT2 prescription items**.

The remaining market is primarily represented by empagliflozin, with substantially smaller shares held by canagliflozin and ertugliflozin.

This demonstrates a highly concentrated competitive market in which dapagliflozin represents the dominant molecule by prescribing volume.

10. Monthly Prescribing Trend

A monthly trend analysis was created covering **January 2025 to June 2026**.

The trend demonstrates that dapagliflozin prescribing increased materially across the period and strengthened its position as the largest molecule in the category.

Empagliflozin remained the second-largest molecule but showed a comparatively flatter prescribing pattern during the later months of the analytical period.

Canagliflozin and ertugliflozin represented considerably smaller volumes throughout the period.

The monthly view is important because annual totals alone can hide changes in market momentum. Observing the monthly trajectory makes it possible to distinguish sustained expansion from temporary changes in volume.

11. Molecule-Level Year-on-Year Growth

The dashboard also compares the growth of each molecule between January–June 2025 and January–June 2026.

Dapagliflozin recorded strong positive growth during the period, while several competing molecules experienced declining prescription volumes.

This demonstrates that the expansion of the SGLT2 category is not evenly distributed between competitors.

A product can therefore experience very different performance from the category as a whole.

The distinction between market growth and product growth is commercially important because a molecule may increase in absolute volume while losing relative competitive position, or alternatively gain market share even within a rapidly expanding category.

12. Contribution to Market Growth

A growth-contribution measure was created to identify how much each molecule contributed to the net expansion of the SGLT2 category.

The calculation compares each molecule's change in prescription volume with the total change in market volume.

Dapagliflozin contributed **more than 100% of net category growth** during 2026 YTD.

This can occur because contribution is calculated against net market growth. While dapagliflozin added substantial prescribing volume, several competing molecules declined. Those declines offset part of the incremental volume generated by dapagliflozin.

The result therefore provides a stronger commercial insight than simple growth percentages.

The category expanded overall, but the expansion was highly concentrated in one molecule rather than being broadly shared across competitors.

13. Prescribing Cost Analysis

The executive dashboard also includes Net Ingredient Cost.

Total NIC for 2026 YTD is approximately **£9.68 billion within the analysed data context**.

However, this figure must not be interpreted as pharmaceutical manufacturer revenue.

Net Ingredient Cost represents a recorded NHS prescribing cost measure and is not equivalent to manufacturer sales, realised revenue, profit, or commercial value captured by an individual pharmaceutical company.

The metric is therefore used as a prescribing expenditure indicator rather than a sales measure.

14. Geographic Opportunity Analysis

Absolute prescribing volume is not appropriate for direct geographic comparison because ICB populations vary considerably.

To address this problem, prescribing activity was normalised using registered GP population.

The principal measure is:

Prescriptions per 1,000 Registered Patients = Geographic Prescription Items / Registered Population × 1,000

Only prescribing activity associated with practices that could be matched to a valid registered-population denominator was included in this geographic rate.

Special or unidentified prescribing codes that did not represent standard GP practices were retained in overall national market analysis but excluded from population-normalised geographic calculations.

This prevents the numerator from including prescribing activity for which no valid population denominator exists.

15. England Prescribing Benchmark

For 2026 YTD, the geographic analysis indicates approximately **144 SGLT2 prescription items per 1,000 registered patients** across the matched England practice population.

This value serves as the national benchmark for comparing utilisation across Integrated Care Boards.

Individual ICBs can therefore be evaluated relative to the overall prescribing rate rather than through raw volume alone.

16. Market Development Index

A Market Development Index was developed to simplify geographic benchmarking.

The measure is calculated as:

$$\text{MDI} = \text{Local Prescriptions per 1,000} / \text{England Prescriptions per 1,000} \times 100$$

An MDI of **100** represents utilisation equivalent to the England benchmark.

An MDI above 100 indicates prescribing utilisation above the national benchmark, while an MDI below 100 indicates lower relative utilisation.

For example, an ICB with an MDI of 120 would have an SGLT2 prescribing rate approximately 20% above the national benchmark.

An ICB with an MDI of 80 would have utilisation approximately 20% below the benchmark.

The MDI provides a simple commercial screening metric, but it should not be interpreted as a direct measure of unmet demand.

17. Geographic Growth

The project also measures year-on-year prescribing growth at ICB level.

Across the matched geographic population, SGLT2 prescribing increased by approximately **20.7% in 2026 YTD compared with the equivalent period in 2025.**

Individual ICBs were then compared with this England growth benchmark.

This creates two complementary dimensions of geographic performance:

Current utilisation, represented by the Market Development Index.

Growth momentum, represented by ICB year-on-year prescribing growth.

Using both metrics avoids interpreting low utilisation in isolation.

18. Geographic Opportunity Segmentation

Each ICB was segmented according to its Market Development Index and growth relative to the England benchmark.

Four commercial segments were created.

Priority Market

These ICBs have:

MDI $\geq$ 100 and growth equal to or above England growth.

They combine strong existing utilisation with strong growth momentum.

Commercially, these markets may warrant continued investment, competitive defence, and efforts to protect existing position.

Emerging Opportunity

These ICBs have:

MDI < 100 and growth equal to or above England growth.

Current utilisation remains below the national benchmark, but growth momentum is relatively strong.

These areas may indicate developing adoption and therefore represent useful candidates for further commercial investigation.

Mature Market

These ICBs have:

MDI $\geq$ 100 and growth below England growth.

Utilisation is already relatively high, but expansion is slower than the wider market.

The commercial focus in these areas may therefore be more closely associated with retention and competitive defence than aggressive expansion.

White Space / Investigate

These ICBs have:

MDI < 100 and growth below England growth.

These areas show both lower utilisation and lower growth.

However, they should not automatically be interpreted as commercial opportunities.

Low utilisation could reflect genuine barriers such as lower eligible-patient prevalence, demographic differences, clinical policy, local treatment pathways, prescribing behaviour, or patient eligibility.

These markets therefore require additional investigation before commercial resources are committed.

19. Commercial Opportunity Map

A geographic opportunity scatterplot was created using:

X-axis: Market Development Index

Y-axis: ICB YoY Growth

Bubble size: Geographic Prescribing Volume

A vertical reference line at MDI = 100 represents the England utilisation benchmark.

A horizontal reference line at approximately 20.7% growth represents England market growth.

The resulting four-quadrant framework visually separates Priority Markets, Emerging Opportunities, Mature Markets, and White Space / Investigation areas.

This enables commercial decision-makers to identify both the relative size of each geographic market and the relationship between current adoption and growth momentum.

20. Key Commercial Insights

Category Growth

SGLT2 prescribing increased by approximately **20% year-on-year during 2026 YTD compared with January–June 2025.**

The category therefore remains in a period of meaningful prescribing expansion.

Market Leadership

Dapagliflozin holds approximately **73% market share by prescription items in 2026 YTD**, making it the clear volume leader within the analysed SGLT2 category.

Growth Concentration

Dapagliflozin generated more than **100% of net incremental category growth.**

This indicates that the overall increase in SGLT2 prescribing was highly concentrated, while declines in several competing molecules offset part of dapagliflozin's incremental volume.

Geographic Variation

SGLT2 utilisation varies materially across Integrated Care Boards.

The difference between high- and low-MDI markets demonstrates that prescribing adoption is not geographically uniform across England.

This variation creates a basis for more targeted commercial investigation.

21. Commercial Recommendations

Protect High-Value Markets

High-MDI, high-growth ICBs should be treated as important commercial markets.

These geographies already demonstrate strong relative utilisation while continuing to expand at or above the market benchmark.

Commercial priorities may therefore include maintaining competitive position, supporting existing momentum, and monitoring potential competitive threats.

Investigate Emerging Opportunities

Low-MDI ICBs that are growing faster than the England benchmark represent particularly interesting markets for further investigation.

Their prescribing utilisation remains relatively underdeveloped compared with the national benchmark, but their growth momentum suggests that adoption may be accelerating.

These markets may therefore warrant deeper analysis to understand the drivers of growth and whether additional commercial focus is justified.

Diagnose White Space Before Investing

Low-MDI, below-market-growth ICBs should not automatically be labelled as underpenetrated commercial opportunities.

Further analysis should investigate local prevalence, demographics, clinical policy, healthcare pathways, patient eligibility, and prescribing behaviour.

Commercial investment should therefore follow diagnostic analysis rather than rely solely on low utilisation.

22. Limitations

Several important limitations affect the interpretation of the analysis.

Prescription Items Are Not Patients

One patient may receive multiple prescriptions during the analytical period.

Prescription items therefore represent prescribing activity rather than unique patient numbers.

It would be incorrect to interpret nine million prescription items as nine million patients.

Prescribing Cost Is Not Pharmaceutical Revenue

NIC and Actual Cost are NHS prescribing expenditure measures.

They do not directly represent the revenue received by pharmaceutical manufacturers.

Commercial statements should therefore refer to recorded prescribing cost rather than product sales or company revenue.

Volume Market Share Is Not Revenue Market Share

The market-share analysis is based on prescription items.

Products may have different prices, pack sizes, strengths, or commercial arrangements.

Prescription-volume share therefore does not necessarily correspond to value or revenue share.

Registered Population Is a Proxy

Registered GP population improves geographic comparison, but it is not equivalent to the population clinically eligible for SGLT2 treatment.

True commercial opportunity would ideally account for the prevalence of relevant conditions such as type 2 diabetes, chronic kidney disease, and heart failure.

Indication Cannot Be Directly Observed

SGLT2 medicines may be prescribed for multiple clinical indications.

The dataset identifies the medicine prescribed but does not necessarily reveal whether the treatment was intended for diabetes, CKD, heart failure, or another eligible use.

Clinical interpretation must therefore remain cautious.

Organisational Matching Is Imperfect

Some prescribing records relate to unidentified doctors, special organisational codes, or practices that could not be matched to the registered-population dataset.

These records were retained when calculating overall market activity but excluded from population-normalised geographic rates where no valid denominator was available.

Correlation Does Not Establish Causality

Changes in prescribing may coincide with clinical guidelines, policy changes, competitive activity, or other market developments.

The prescribing dataset alone cannot establish that any individual factor caused the observed changes.

23. Future Improvements

The geographic opportunity model could be strengthened by incorporating disease prevalence for type 2 diabetes, chronic kidney disease, and heart failure.

Age distribution could also improve the denominator because SGLT2 eligibility and disease prevalence vary substantially between demographic groups.

Additional variables such as deprivation, prescribing-policy differences, healthcare access, local formularies, and clinical pathway information could provide additional explanation for geographic variation.

Using monthly registered-patient populations throughout the complete January 2025 to June 2026 period would also improve temporal alignment between prescribing activity and the population denominator.

Future analysis could also investigate product presentation, strength, prescribing cost per item, regional competitive-share movement, and changes around major clinical or policy developments.

24. Dashboard Structure

The final Power BI solution consists of three analytical pages.

Executive Commercial Overview

The first page provides a high-level view of market performance.

It includes total SGLT2 prescription items, market YoY growth, Net Ingredient Cost, leading molecule, leading molecule market share, monthly prescribing trends, current market share, molecule-level YoY growth, and contribution to market growth.

The purpose of the page is to answer:

What is happening in the SGLT2 market overall, and which products are driving the change?

Geographic Opportunity

The second page evaluates variation between Integrated Care Boards.

It includes the national prescriptions-per-1,000 benchmark, England geographic growth, matched prescribing volume, highest and lowest utilisation markets, the ICB opportunity scatterplot, Market Development Index, geographic YoY growth, and ICB segmentation.

The purpose of the page is to answer:

Where is utilisation relatively high or low, and which markets warrant further commercial investigation?

Commercial Insights, Recommendations & Limitations

The final page translates analytical findings into managerial interpretation.

It summarises key commercial insights, recommended areas of action, methodological limitations, and potential improvements.

The purpose is to ensure that the project moves beyond descriptive visualisation and demonstrates the ability to translate data into commercial decision support.

25. Skills Demonstrated

The project demonstrates the integration of technical analytics, commercial thinking, and information-management skills.

From a Power BI perspective, the project includes Power Query transformation, multi-file data ingestion, schema harmonisation, star-schema modelling, DAX, time intelligence, measures, population-normalised metrics, conditional formatting, ranking, reference lines, filtering, and interactive visualisation.

From a commercial analytics perspective, the project demonstrates market sizing, competitive-share analysis, year-on-year growth analysis, growth decomposition, geographic benchmarking, market segmentation, opportunity screening, and commercial recommendation development.

From an information-systems perspective, the project demonstrates the integration of heterogeneous public datasets, data-quality assessment, data modelling, transformation of

transactional records into managerial information, and the design of a decision-support system.

The project also demonstrates analytical judgement by explicitly distinguishing what the data can support from conclusions that would require additional evidence.

26. Conclusion

The analysis demonstrates how publicly available NHS prescribing data can be transformed into commercially relevant pharmaceutical market intelligence.

The SGLT2 category continued to expand during 2026 YTD, with approximately 20% year-on-year growth compared with the equivalent period in 2025.

Dapagliflozin maintained a dominant competitive position, accounting for approximately 73% of prescription items and driving the majority of incremental category growth.

At the same time, substantial variation exists across England in both prescribing utilisation and market growth.

The Market Development Index and geographic segmentation framework provide a structured method for identifying markets that may warrant commercial prioritisation or further investigation.

However, the project also demonstrates the importance of cautious interpretation. Prescribing activity is not equivalent to patient numbers or pharmaceutical revenue, and lower geographic utilisation cannot automatically be interpreted as unmet commercial demand.

The dashboard should therefore be understood as **a commercial opportunity-screening and decision-support tool**, rather than a definitive model of pharmaceutical demand.

By combining market growth, competitive dynamics, geographic benchmarking, and explicit consideration of analytical limitations, the project demonstrates how healthcare data can support more structured and evidence-based pharmaceutical commercial decision-making.